# Converting R Code to Equation

https://gemini.google.com/app/16c446370419cf17

*User prompt: P_new <- 1 - exp(-(2*N_0*mu*(s-r))/(r)) This is a code-snipped in R. Give me the equation in a nice way.*

Response:

$$P_{\text{new}} = 1 - \exp\left(-\frac{2N_0\mu(s-r)}{r}\right)$$

Or written in standard exponential form:

$$P_{\text{new}} = 1 - e^{-\frac{2N_0\mu(s-r)}{r}}$$

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*User prompt: Given that i have two age classes and for both of them a reproduction- and a survival rate. For the juveniles the reproduction is 0 and the survival is given by S_j. For the adults its R_a and S_a. Do i get the growth rate of the population by calculating the biggest eigenvalue of the corresponding leslie-matrix?*

Response: To determine whether the population growth rate corresponds to the largest eigenvalue, consider the formulation of the transition model and its mathematical properties.

### 1. Constructing the Projection Matrix

Let the population vector at time $t$ be:

$$\mathbf{n}_t = \begin{bmatrix} J_t \\ A_t \end{bmatrix}$$

where $J_t$ is the number of juveniles and $A_t$ is the number of adults.

Depending on the demographic census timing (pre-breeding vs. post-breeding), the projection matrix $\mathbf{M}$ is defined such that $\mathbf{n}_{t+1} = \mathbf{M}\mathbf{n}_t$:

- **First row (juvenile recruitment):** Juveniles produce $0$ offspring. Adults produce new juveniles with effective fecundity $F_a$ (which combines adult reproduction $R_a$ and survival depending on census timing).
- **Second row (adult recruitment/retention):** Juveniles survive to become adults with probability $S_j$. Existing adults survive with probability $S_a$.

$$\mathbf{M} = \begin{bmatrix} 0 & F_a \\ S_j & S_a \end{bmatrix}$$

*(Note: Because adults can survive and remain in the adult class via $S_a > 0$, this is technically a **Lefkovitch stage-structured matrix**, rather than a strict age-class Leslie matrix where individuals must advance to an older age class each step).*

### 2. The Dominant Eigenvalue

The characteristic equation $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$ is:

$$\det\begin{bmatrix} -\lambda & F_a \\ S_j & S_a - \lambda \end{bmatrix} = (-\lambda)(S_a - \lambda) - S_j F_a = 0$$

$$\lambda^2 - S_a\lambda - S_j F_a = 0$$

Using the quadratic formula:

$$\lambda = \frac{S_a \pm \sqrt{S_a^2 + 4S_j F_a}}{2}$$

Since $S_a \geq 0$, $S_j > 0$, and $F_a > 0$, the discriminant is strictly positive, yielding one positive real root:

$$\lambda_1 = \frac{S_a + \sqrt{S_a^2 + 4S_j F_a}}{2}$$

Because $\sqrt{S_a^2 + 4S_j F_a} > S_a$, we have $\lambda_1 > |\lambda_2|$.

By the **Perron-Frobenius theorem**, because $\mathbf{M}$ is non-negative and primitive (provided $F_a > 0$ and $S_j > 0$), there exists a unique, real, positive **dominant eigenvalue** $\lambda_1$ that is strictly larger in absolute magnitude than all other eigenvalues.

In the long run ($t \to \infty$):

- $\mathbf{n}_t \approx c\lambda_1^t \mathbf{v}_1$, where $\mathbf{v}_1$ is the right eigenvector (the stable stage distribution).
- The finite rate of population increase per time step is $\lambda = \lambda_1$.
- The intrinsic rate of natural increase is $r = \ln(\lambda_1)$.

### Verdict

**Yes**, calculating the largest (dominant) eigenvalue of the matrix gives the asymptotic finite growth rate ($\lambda$) of the population.

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*User prompt: Don't change anything in the model!!! I have now a table which contains both the estimated rescue probability calculated by the formula and the one calculated by the actual simulations. I still have some negative values for the expected rescue probability. These rows must be removed before plotting. Then i want to plot the rescue probability on the y-axis and the lambda of the mutants on the x-axis for both the estimated and the "actual" rescue probability. Give me a code-snipped to create such a plot. I think i already used GGplot in the document so no new packages should have to be installed and stuff.*

Response:
```
# 1. Calculate mutant lambda (growth_rate_WT + s)
summary_table$lambda_mutant <- growth_rate_WT + summary_table$selection_coefficient

# 2. Filter out negative estimated rescue probabilities and reshape for plotting
plot_data <- summary_table %>%
  filter(rescue_probability_equation >= 0) %>%
  rename(Simulated = `resque probability`, Formula = rescue_probability_equation) %>%
  pivot_longer(
    cols = c(Simulated, Formula),
    names_to = "Source",
    values_to = "Rescue_Probability"
  )

# 3. Create the plot
ggplot(plot_data, aes(x = lambda_mutant, y = Rescue_Probability, color = Source)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(se = FALSE, method = "loess", span = 0.8) +
  theme_minimal() +
  labs(
    title = "Evolutionary Rescue Probability vs. Mutant Growth Rate",
    x = expression(paste("Mutant Growth Rate (", lambda[mut], ")")),
    y = "Rescue Probability",
    color = "Type"
  ) +
  scale_color_manual(values = c("Simulated" = "steelblue", "Formula" = "darkorange")) +
  ylim(0, 1)
```

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*User prompt: You didn't use the names of my document, right? Do you even have access to it? Tell me the name of the first function that i defined in the model to test that.*

Response: Yes, I have access to your document (`evolutionary-rescue-basic-discrete-time.Rmd`).

The first function defined in your model is:

`simulate_one_gen`

(defined right under `# Simulate one generation` to simulate one time-step using Poisson draws for juveniles and adults).

The previous snippet did use your exact column and variable names, including:

- `summary_table`
- `` `resque probability` `` (matching the spelling in your script)
- `rescue_probability_equation`
- `selection_coefficient`

- growth_rate_WT

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*User prompt: But it gives me the following error message: Fehler in summary_table %>% filter(rescue_probability_equation >= 0) %>% : konnte Funktion "%>%" nicht finden*

Response: Dieser Fehler tritt auf, weil der Pipe-Operator %>% aus dem Paket **dplyr** (bzw. magrittr) stammt und in deiner aktuellen R-Session in diesem Chunk noch nicht geladen ist.

Obwohl die Pakete in einem früheren Code-Chunk geladen wurden, müssen sie in der aktiven Sitzung verfügbar sein. Setze einfach library(dplyr) und library(tidyr) an den Anfang des Blocks:

```
library(dplyr)
library(tidyr)
library(ggplot2)

# 1. Lambda der Mutanten berechnen (growth_rate_WT + s)
summary_table$lambda_mutant <- growth_rate_WT + summary_table$selection_coefficient

# 2. Negative Werte herausfiltern und Daten für ggplot umformen
plot_data <- summary_table %>%
  filter(rescue_probability_equation >= 0) %>%
  rename(Simulated = `resque probability`, Formula = rescue_probability_equation) %>%
  pivot_longer(
    cols = c(Simulated, Formula),
    names_to = "Source",
    values_to = "Rescue_Probability"
  )

# 3. Plot erstellen
ggplot(plot_data, aes(x = lambda_mutant, y = Rescue_Probability, color = Source)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(se = FALSE, method = "loess", span = 0.8) +
  theme_minimal() +
  labs(
    title = "Evolutionary Rescue Probability vs. Mutant Growth Rate",
    x = expression(paste("Mutant Growth Rate (", lambda[mut], ")")),
    y = "Rescue Probability",
    color = "Type"
  ) +
  scale_color_manual(values = c("Simulated" = "steelblue", "Formula" = "darkorange")) +
  ylim(0, 1)
```

### Alternative ohne Pipe-Operator (reines Base R + ggplot2)

Falls du dplyr und tidyr gar nicht erst laden möchtest, kannst du das Filtern direkt in Base R vornehmen und die beiden Kurven als separate Schichten einzeichnen:

```
library(ggplot2)

# 1. Lambda der Mutanten berechnen
summary_table$lambda_mutant <- growth_rate_WT + summary_table$selection_coefficient

# 2. Zeilen mit negativen Werten in Base R filtern
plot_df <- summary_table[summary_table$rescue_probability_equation >= 0, ]

# 3. Beide Wahrscheinlichkeiten direkt plotten
ggplot(plot_df, aes(x = lambda_mutant)) +
  geom_point(aes(y = `resque probability`, color = "Simulated"), alpha = 0.7, size = 2) +
  geom_smooth(aes(y = `resque probability`, color = "Simulated"), method = "loess", se = FALSE, span = 0.8) +
  geom_point(aes(y = rescue_probability_equation, color = "Formula"), alpha = 0.7, size = 2) +
  geom_smooth(aes(y = rescue_probability_equation, color = "Formula"), method = "loess", se = FALSE, span = 0.8) +
  theme_minimal() +
  labs(
    title = "Evolutionary Rescue Probability vs. Mutant Growth Rate",
    x = expression(paste("Mutant Growth Rate (", lambda[mut], ")")),
    y = "Rescue Probability",
    color = "Type"
  ) +
  scale_color_manual(values = c("Simulated" = "steelblue", "Formula" = "darkorange")) +
  ylim(0, 1)
```

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